

Generation Interconnection Request System Impact Study Report

For Queue Project Q17-0881

Lookout 138 kV
May 20, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0881. The Interconnection Customer proposes a 188.0 MW solar photovoltaic facility interconnecting at the Lookout 138 kV substation in the Virginia area. The request was received on October 17, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 314 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$108,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Falu 1334 / Burrows Crossing Road 138 KV: 105.8% loading (440 MVA) following the outage of the Falu 1334 / Burrows Crossing Rd 138 KV; DFAX = 6.0%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Falu 1334 / Burrows Crossing Road 138 KV	Reconductor line	\$321,000	34% (\$108,000)
	Total allocated to project		\$108,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.